

Siddarth Akalankam

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EDUCATION

Michigan State University

Bachelor of Science in Computer Science, Minor in Business

GPA: 3.76

East Lansing, MI

Aug. 2022 – May 2026

EXPERIENCE

Amazon

Capstone Software Engineer

East Lansing, MI

Jan. 2026 – Apr. 2026

- Architected a unified observability platform automating anomaly detection across **AWS services** deployed in multiple global regions, saving **2+ hours per day** of manual monitoring for engineering teams.
- Designed a serverless **ETL pipeline** using **EventBridge** and **Lambda** to extract, transform, and load scheduled service metrics into **DynamoDB** for downstream processing.
- Enabled multi-modal detection by integrating **SageMaker** for time-series metrics and **Rekognition Custom Labels** for screenshot analysis, automating severity and priority ranking across both data sources.
- Built an anomaly triage system with direct **CloudWatch** deep links to anomaly timestamps, reducing investigation scope from **10,000+** logs to roughly **50** per incident.
- Enabled secure multi-account observability by implementing **IAM role assumption** for cross-account metric ingestion without credential sharing.

Altair

Software Engineering Intern

Troy, MI

May 2025 – Aug. 2025

- Shipped a production **REST API** on **Azure** that abstracted all infrastructure management for data scientists, enabling one-click **ML model** deployment across cloud and edge.
- Cut inference latency from **4s to 100ms** by building a **C++ TensorRT** pipeline for **NVIDIA Jetson** devices, migrating cloud inference from CPU to GPU, and resolving networking bottlenecks.
- Integrated the platform into Altair's existing microservice architecture by containerizing the API with **Docker** and deploying it to **Kubernetes** with custom YAML configurations.

RecycleMe

ML Software Engineering Intern

East Lansing, MI

Jan. 2024 – Apr. 2024

- Deployed a **PyTorch ResNet-101** model on **AWS SageMaker** achieving **92% accuracy** across 6 material categories, fully replacing manual classification.
- Reduced external API calls by shifting inference on-device within the **Flutter/Firebase** app, improving real-time response for end users.

PROJECTS

FlashML | *Next.js, FastAPI, ONNX Runtime, AWS ECS, AWS S3, PostgreSQL, RunPod Serverless*

- Built **FlashML**, a platform that enables developers to deploy machine learning models as **production-ready APIs** through a single upload, without managing infrastructure, GPUs, or containers.
- Engineered an automated deployment pipeline for **model validation, versioning, rollback, and endpoint generation**, supporting CPU and GPU inference through **ONNX Runtime**.

FinForge AI | *React, TypeScript, FastAPI, Plaid, Claude API, AWS Lambda, AWS SQS, PostgreSQL, Pinecone*

- Built **FinForge**, a personal finance AI agent that lives in the user's inbox, syncing live banking data through **Plaid**, generating personalized financial briefings, and delivering them through scheduled email workflows.
- Engineered an agentic retrieval system that dynamically retrieves financial data from **PostgreSQL**, long-term memory from **Pinecone**, and external tools to answer follow-up financial questions through email conversations.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, C/C++, SQL, Java

Cloud & Infra: AWS, Azure, GCP, Docker, Kubernetes, Terraform, CDK, CI/CD

Databases & Backend: PostgreSQL, DynamoDB, NoSQL, REST APIs, gRPC, Spring Boot, FastAPI

AI/ML & Developer Tools: PyTorch, ONNX, SageMaker, RAG, Cursor, Codex, Claude Code